

# WASP-10b

R-0684

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-10\_241006 · created 2026-07-27T19:58:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460589.7068 +/- 0.0032 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.173 +/- 0.018                 |
| Transit depth (Rp/Rs)^2                           | 3.01 +/- 0.64 [%]               |
| Orbital Inclination (inc)                         | 89.91 +/- 0.81                  |
| Airmass coefficient 1 (a1)                        | 1.021 +/- 0.016                 |
| Airmass coefficient 2 (a2)                        | -0.02 +/- 0.046                 |
| Scatter in the residuals of the lightcurve fit is | 1.53 %                          |
| Best Comparison Star                              | #1 - [344, 342]                 |
| Optimal Aperture                                  | 3.65                            |
| Optimal Annulus                                   | 10.86                           |
| Transit Duration (day)                            | 0.0948 +/- 0.0022               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.173 ± 0.018 vs 0.1592 (deviation 0.77σ) |
| Duration fit vs published | 0.0948 d vs 0.0946 d (deviation 0.1σ)     |
| Mid-time O–C              | 2.8 min                                   |
| Depth SNR                 | 9.6                                       |
| Residual scatter          | 1.53 %                                    |

## Light curve

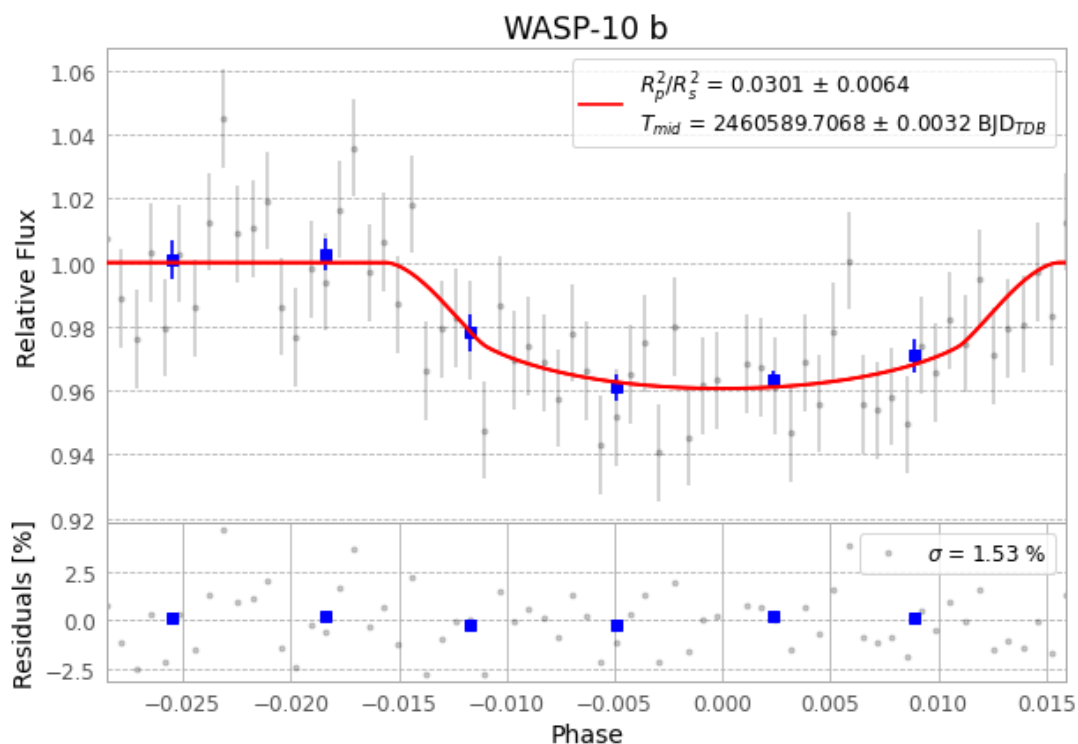


Figure 1 — FinalLightCurve\_WASP-10 b\_2024-10-05.png (SHA-256 f7d2b9498fee57c6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 216], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 a1e73b354f80e8dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 62f7813

## Data provenance

67 FITS frames (44 MB), WASP-10241006024216.FITS → WASP-10241006060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-241006.log (4d1e164da5d8...), FinalLightCurve\_WASP-10 b\_2024-10-05.png (f7d2b9498fee...), FinalLightCurve\_WASP-10 b\_2024-10-05.csv (a5625d5dd1e9...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:58Z.